

$$\underline{x}(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}; \underline{A} = (a_{ij})$$

$i, j = 1, \dots, n$

$$\left. \begin{aligned} \frac{d}{dt} \underline{x}(t) &= \underline{A} \underline{x}(t) \\ \underline{x}(t=0) &= \underline{y}_0 \end{aligned} \right\}$$

$$\underline{x}(t) = e^{\underline{A} \cdot t} \underline{y}_0$$

$$= \sum_{k=0}^{\infty} \underline{A}^k t^k \cdot \frac{1}{k!} \underline{y}_0$$

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$$\underline{A} = \underline{P} \underline{D} \underline{P}^{-1}$$

$$\underline{P} = \begin{pmatrix} \underline{x}_1 & \underline{x}_2 & \underline{x}_3 & \dots & \underline{x}_n \end{pmatrix}$$

column vector
= eigenvector

$$\underline{D} = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$$

$$e^{\underline{A}} = \sum_{k=0}^{\infty} \underline{A}^k / k! =$$

$$= \sum_{k=0}^{\infty} (\underline{P} \underline{D} \underline{P}^{-1})^k / k!$$

$$= \sum_{k=0}^{\infty} (\underline{P} \underline{D} \underline{P}^{-1})^k / k!$$

-3-

$$\begin{aligned} A^2 &= (P D P^{-1})(P D P^{-1}) \\ &= P D^2 P^{-1} \end{aligned}$$

$$\begin{aligned} A^3 &= (P D P^{-1})^3 \\ &= P D P^{-1} P D P^{-1} P D P^{-1} \\ &= P \cdot D^3 P^{-1} \end{aligned}$$

$$A^k = P D^k P^{-1}$$

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$$\underline{\underline{A}} = \underline{\underline{P}} \underline{\underline{D}} \underline{\underline{P}}^{-1}$$

$$e^{\underline{\underline{A}}} = e^{\underline{\underline{P}} \underline{\underline{D}} \underline{\underline{P}}^{-1}}$$

$$= \sum_{k=0}^{\infty} (\underline{\underline{P}} \underline{\underline{D}} \underline{\underline{P}}^{-1})^k / k!$$

$$= \underline{\underline{P}} \sum_{k=0}^{\infty} \underline{\underline{D}}^k / k! \cdot \underline{\underline{P}}^{-1}$$

$$\boxed{D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} = \underline{\underline{P}} e^{\underline{\underline{D}}} \underline{\underline{P}}^{-1}}$$

$$D^2 = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} = \begin{pmatrix} \lambda_1^2 & & \\ & \ddots & \\ & & \lambda_n^2 \end{pmatrix}$$

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$$\Theta^k = \begin{pmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{pmatrix}$$

$$\rho^k = \sum_{r=0}^k D^r / r!$$

$$= \begin{pmatrix} \sum_{r=0}^k \lambda_1^r / r! & \emptyset & \\ \emptyset & \sum_{r=0}^k \lambda_2^r / r! & \\ & & \ddots \\ \emptyset & & & \sum_{r=0}^k \lambda_n^r / r! \end{pmatrix}$$

$$= \begin{pmatrix} e^{\tau_1} & & \emptyset \\ \emptyset & e^{\tau_2} & \\ & & \ddots \\ \emptyset & & & e^{\tau_n} \end{pmatrix}$$

6-

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{x}(t=0) = \underline{y}$$

$$A = P D P^{-1}$$

$$x(t) = e^{\underline{A}t}$$

$$= P e^{\underline{D}t} P^{-1} \underline{y} =$$

$$= P \begin{pmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{pmatrix} P^{-1} \underline{y}$$

$$\underline{x} = P \underline{z}$$

$$\dot{\underline{x}} = P \dot{\underline{z}} = \underline{A} \underline{x} = P D P^{-1} P \underline{z}$$

$$\dot{\underline{z}} = D \underline{z}$$

$$= P D \underline{z}$$

$$\frac{d}{dt} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

$$\underline{x}(t=0) = \begin{pmatrix} x(t=0) \\ y(t=0) \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

Let $\underline{x}_1, \underline{x}_2$ and λ_1, λ_2 satisfy

$$\underline{A} \underline{x}_1 = \lambda_1 \underline{x}_1$$

$$\underline{A} \underline{x}_2 = \lambda_2 \underline{x}_2$$

$$\underline{x}_1 = \begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix}$$

$$\underline{x}_2 = \begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix}$$

$$P = \begin{pmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{pmatrix}$$

-8-

$$\underline{x} = \underline{p} \cdot \underline{z}$$

$$\dot{\underline{z}} = \underline{D} \cdot \underline{z}$$

$$\underline{z}(t) = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix} \begin{pmatrix} z_{10} \\ z_{20} \end{pmatrix}$$

$$\underline{z}(t=0) = \underline{z}_0 = \begin{pmatrix} z_{10} \\ z_{20} \end{pmatrix}$$

$$\begin{aligned} \underline{x}(t) &= \underline{p} \underline{z} = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix} \begin{pmatrix} z_{10} \\ z_{20} \end{pmatrix} \\ &= \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \begin{pmatrix} z_{10} e^{\lambda_1 t} \\ z_{20} e^{\lambda_2 t} \end{pmatrix} \end{aligned}$$

-9-

$$\underline{x}(t) = \begin{pmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \end{pmatrix} \begin{pmatrix} z_{10} e^{\lambda_1 t} \\ z_{20} e^{\lambda_2 t} \end{pmatrix}$$

$$= \begin{pmatrix} x_{11} z_{10} e^{\lambda_1 t} + x_{21} z_{20} e^{\lambda_2 t} \\ x_{12} z_{10} e^{\lambda_1 t} + x_{22} z_{20} e^{\lambda_2 t} \end{pmatrix}$$

$$= \begin{pmatrix} x_{11} \\ x_{12} \end{pmatrix} z_{10} e^{\lambda_1 t} + \begin{pmatrix} x_{21} \\ x_{22} \end{pmatrix} z_{20} e^{\lambda_2 t}$$

$$= z_{10} \underline{x}_1 e^{\lambda_1 t} + z_{20} \underline{x}_2 e^{\lambda_2 t}$$

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{x}(t=0) = \underline{y}$$

$$\text{is } \underline{x}(t) = \sum_{k=1}^n c_k e^{\lambda_k t} \underline{x}_k$$

-10-

Solve $\underline{\dot{x}} = \underline{A} \underline{x}$

with $\underline{A} = \begin{pmatrix} d & 0 \\ 0 & -1 \end{pmatrix}$

for $-\infty < d < \infty$

and plot phase portrait

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$$A = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\lambda_{1,2} = 2; -1$$

$$\lambda_1 = 2; \lambda_2 = -1$$

$$\underline{x}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \underline{x}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{aligned} \underline{x}(t) &= c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-t} \\ &= \begin{pmatrix} x_0 e^{2t} \\ y_0 e^{-t} \end{pmatrix} \end{aligned}$$

$$x(t) = x_0 e^{2t}$$

$$y(t) = y_0 e^{-t}$$

$$e^t = y_0 / y(t)$$

$$x(t) = x_0 \left(\frac{y_0}{y(t)} \right)^2$$

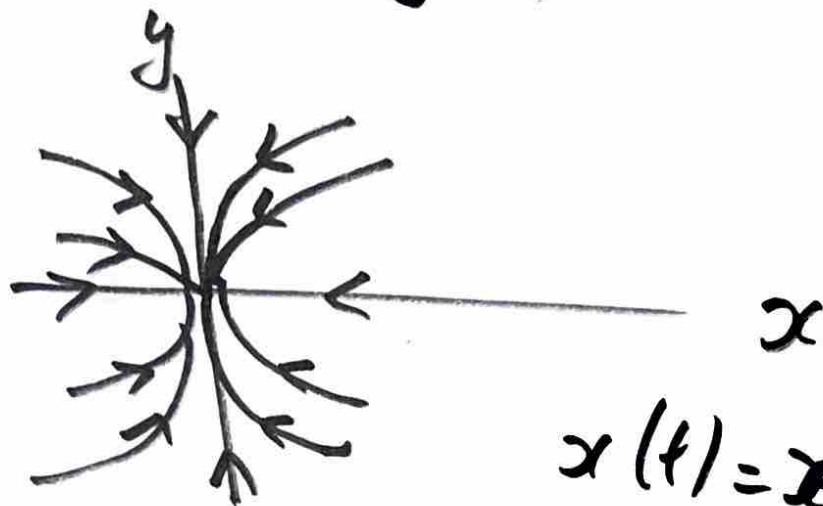
$$x(t) = C y(t)^{-2}$$

$$x(t) = C y(t)^{-2}$$

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$$\lambda < -1$$

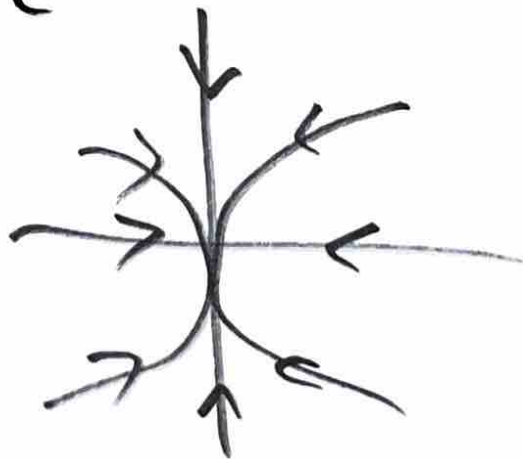
$$\underline{\lambda = -2} \quad x(t) = (y^2(t))$$



$$x(t) = x_0 e^{-2t}$$

$$y(t) = y_0 e^{-t}$$

Stable
node



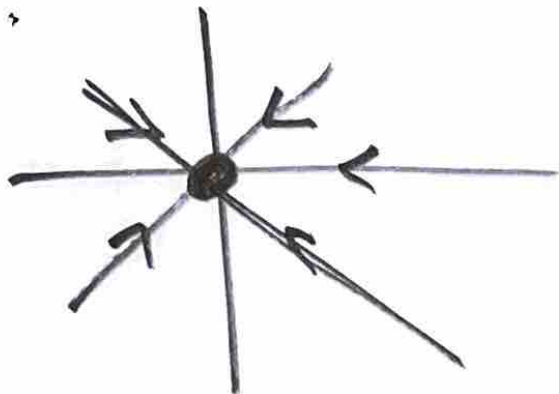
Trajectories
are tangential to
a smaller (in absolute value)

(eigenvalue)

-13-

$$\lambda = -1$$

$$x(t) = C(y(t))^{-\lambda} = C y(t)$$



stable star

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$$\lambda = -2$$

$$\lambda = -2$$

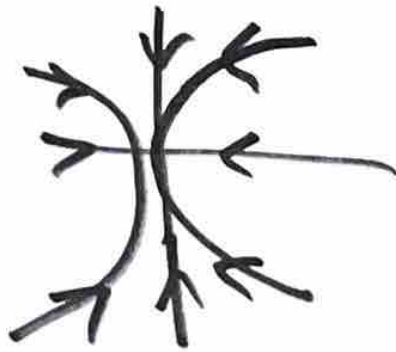
$$x(t) = x_0 e^{-2t}$$

$$y(t) = y_0 e^{-t}$$

$$e^{-t} = y(t) / y_0$$

$$e^{-2t} = (e^{-t})^2 = y^2(t) / y_0^2$$

$$x(t) = x_0 y^2(t) / y_0^2$$



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$$-1 < \alpha < 0$$

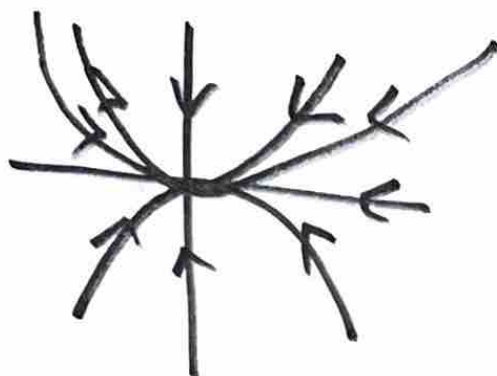
$$\alpha = -\frac{1}{2}$$

$$x(t) = C (y(t))^{-\alpha}$$

$$x(t) = x_0 e^{-\frac{t}{2}}$$

$$y(t) = y_0 e^{-t}$$

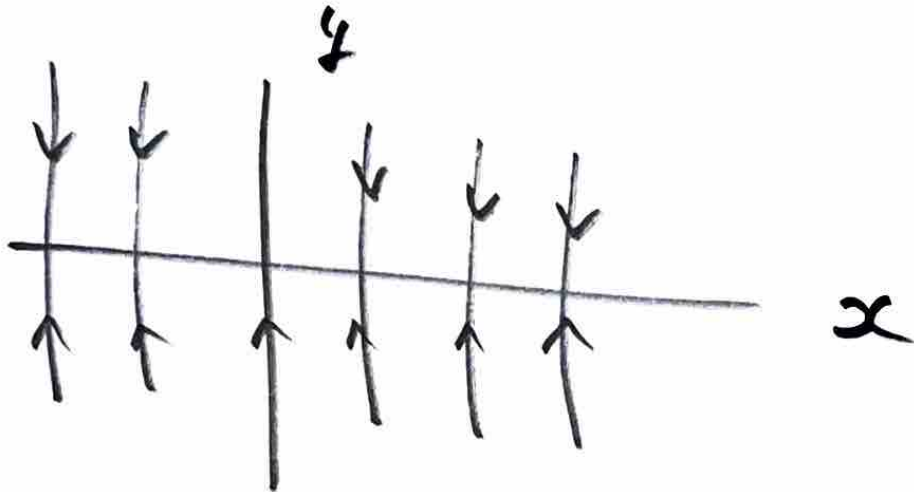
$$x(t) = C \sqrt{y}$$



-16- $\mathcal{L} = \emptyset$

$$x(t) = x_0$$

$$y(t) = y_0 e^{-t}$$



∞ many fixed
points
 $(x_0, 0)$ is a
fixed
point

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$$\alpha > 0$$

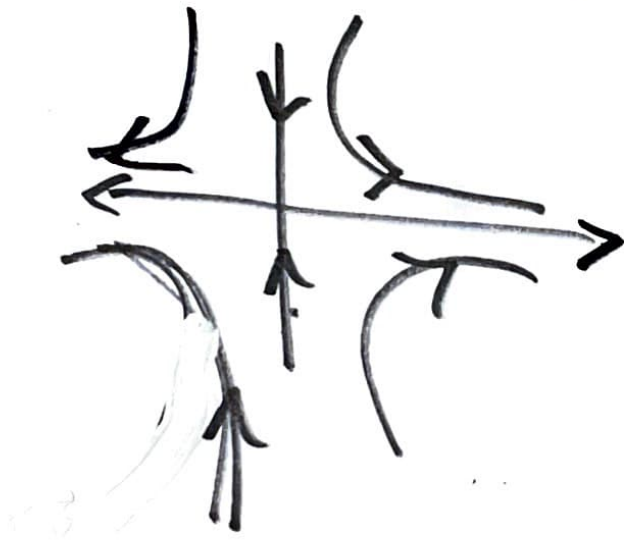
$$x(t) = x_0 e^t$$

$$\alpha = 1$$

$$y(t) = y_0 e^{-t}$$

$$x(t) = C y(t)^{-1}$$

$$x(t) = C / y(t)$$



-18- Example

$$A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$$

$$\frac{d}{dt} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

Solve for $x(t)$, $y(t)$,
plot phase portrait

$$\text{Det}[A - \lambda I] =$$

$$= \text{Det} \begin{bmatrix} 1-\lambda & 1 \\ 4 & -2-\lambda \end{bmatrix}$$

$$= -(1-\lambda)(2+\lambda) - 4$$

$$= (\lambda-1)(\lambda+2) - 4 = \lambda^2 + \lambda - 6$$

$$\lambda_{1,2} = \frac{-1 \pm \sqrt{25}}{2} = -3; 2$$

$$\bullet \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = -3 \begin{pmatrix} a \\ b \end{pmatrix}$$

$$a + b = -3a \Rightarrow b = -4a$$

$$\underline{x}_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$$\bullet \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = 2 \begin{pmatrix} c \\ d \end{pmatrix}$$

$$c + d = 2c, \quad d = c$$

$$x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda_1 = -3; \quad x_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$$\lambda_2 = 2; \quad x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

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$$\underline{x}(t) = C_1 \underline{x}_1 e^{\lambda_1 t} + C_2 \underline{x}_2 e^{\lambda_2 t}$$

$$= C_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-3t} + C_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}$$

